Unistochastic Dynamics: From Discrete Transitions to the Entropic Continuum

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Abstract

This monograph synthesizes Jacob Barandes's Indivisible Stochastic Quantum Mechanics (ISQM), where unistochastic matrices $\Gamma(t) = |U(t)|^2$ govern stochastic transitions in classical-like configuration spaces, with the Relativistic Scalar–Vector Plenum (RSVP), a field-theoretic framework encoding entropic flow. Unistochastic matrices, derived from unitary operators, replicate quantum phenomena such as interference and entanglement without postulating wavefunctions as ontological primitives. By coarse-graining RSVP's field equations, these matrices emerge as discrete integrals of entropic flux, formalized via a functor $\mathfrak{C}_1 : \mathbf{RSVPField} \to \mathbf{UniStoch}$. RSVP completes quantum mechanics by embedding stochastic probabilities in a thermodynamic geometry, contrasting with Bohmian mechanics' deterministic non-locality. Through derived geometry, BV–BRST quantization, semiclassical expansions, renormalization-group flows, and categorical structures, this work constructs a unified entropic mechanics bridging quantum stochasticity, thermodynamic smoothing, and cognitive coherence across scales.

unistochastic matrix, entropy geometry, shifted symplectic structure, stochastic quantization, derived stacks, AKSZ sigma model

Contents

1 Preliminaries: Mathematical Foundations

To ensure rigor, we establish the mathematical prerequisites for unistochastic dynamics and the RSVP framework, addressing functional spaces, probabilistic normalization, and well-posedness.

1.1 Functional Spaces and Well-posedness

The RSVP fields $(\Phi, \mathbf{v}, S)$ are defined on a compact (1+3)-dimensional Lorentzian manifold Σ with boundary $\partial\Sigma$, assumed to lie in Sobolev spaces $H^1(\Sigma)$ for regularity, ensuring square-integrable first derivatives. The field equations are:

$$\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = \sigma(S), \quad \partial_t S = D\nabla^2 S - \mathbf{v} \cdot \nabla S + f(\Phi),$$

where $\sigma(S) = -\kappa S^2$ and $f(\Phi) = \lambda\Phi$ are Lipschitz continuous ($\kappa, \lambda < \infty$), ensuring well-posedness. The total probability is normalized:

$$\int_\Sigma \Phi \, d^3x = 1.$$

Theorem 1 (Existence and Uniqueness)**.** *For initial data $(\Phi_0, \mathbf{v}_0, S_0) \in H^1(\Sigma) \times H^1(\Sigma, \mathbb{R}^3) \times H^1(\Sigma)$, with $\int_\Sigma \Phi_0 \, d^3x = 1$, and $\sigma, f \in C^1(\mathbb{R})$ Lipschitz with constants $|\sigma'(S)| \leq L_\sigma$, $|f'(\Phi)| \leq L_f$, there exists a unique classical solution $(\Phi, \mathbf{v}, S) \in C([0,T]; H^1(\Sigma))$ to the RSVP equations on $[0,T]$ for some $T > 0$. If $\|\nabla \cdot \mathbf{v}\|_\infty$, $\|\sigma\|_{C^1}$, and $\|f\|_{C^1}$ are uniformly bounded, the solution extends globally.*

Proof. The system comprises a hyperbolic equation for Φ and a parabolic equation for S, coupled through $\mathbf{v}$. Define the operator $\mathcal{L}_S = \partial_t - D\nabla^2 + \mathbf{v} \cdot \nabla$. By parabolic PDE theory (9), $\mathcal{L}_S$ generates a strongly continuous semigroup on $H^1(\Sigma)$. For Φ, rewrite the first equation as:

$$\partial_t \Phi = -\nabla \cdot (\Phi \mathbf{v}) + \sigma(S).$$

Treating $\sigma(S)$ as a source term, the integral equation is:

$$\Phi(t) = e^{-\int_0^t \nabla \cdot \mathbf{v} \, ds} \Phi_0 + \int_0^t e^{-\int_s^t \nabla \cdot \mathbf{v} \, du} \sigma(S(s)) \, ds.$$

Lipschitz continuity of $\sigma(S)$ and $f(\Phi)$ ensures a contraction mapping in $C([0,T]; H^1(\Sigma))$ for small T, by the Picard-Lindelöf theorem. Energy estimates for Φ:

$$\frac{d}{dt} \|\Phi\|_{H^1}^2 \leq C(\|\Phi\|_{H^1}^2 + \|\sigma(S)\|_{H^1}^2),$$

and similarly for S, with Grönwall's inequality bounding growth. Global existence follows if $\|\nabla \cdot \mathbf{v}\|_\infty$ is bounded, preventing blow-up. Normalization $\int \Phi \, d^3x = 1$ is preserved, as:

$$\frac{d}{dt} \int \Phi \, d^3x = \int \sigma(S) \, d^3x,$$

and $\sigma(S) = -\kappa S^2 \leq 0$ ensures decay or conservation, consistent with probability. □

Boundary conditions are Dirichlet for S ($S|_{\partial\Sigma} = 0$) and Neumann for Φ ($\nabla\Phi \cdot \mathbf{n}|_{\partial\Sigma} = 0$), ensuring physical consistency.

1.2 Probability and Stochastic Processes

A stochastic process on a finite configuration space $X = \{x_i\}_{i=1}^N$ is defined by a probability vector $\rho \in \mathbb{R}^N$, $\sum_i \rho_i = 1$, evolving via a transition matrix Γ_{ij}. A matrix is doubly stochastic if:

$$\sum_i \Gamma_{ij} = 1, \quad \sum_j \Gamma_{ij} = 1, \quad \Gamma_{ij} \geq 0.$$

It is unistochastic if $\Gamma_{ij} = |U_{ij}|^2$ for some unitary $U \in \mathrm{U}(N)$. Non-Markovianity is defined by history-dependent transitions:

$$\Gamma_{ij}(t|h) = \Pr(x_i(t + \Delta t)|x_j(t), h),$$

where h is the trajectory history, breaking the Chapman–Kolmogorov equation.

1.3 Notation Table

- $\Gamma_{ij}(t)$: Unistochastic transition matrix.
- $\Phi(x, t)$: Entropy density (scalar field).
- $\mathbf{v}(x, t)$: Coherence flow (vector field).
- $S(x, t)$: Entropy potential (scalar field).
- ω_0: 0-shifted symplectic form.
- S_{BV}: Batalin-Vilkovisky action.
- $\mathfrak{C}_1$: Coarse-graining functor.
- $H^1(\Sigma)$: Sobolev space of square-integrable functions with weak derivatives.
- $\mathcal{X} = \mathbf{T}^*[0]X$: Cotangent bundle of configuration space X.

2 Introduction: The Need for a Coherent Substrate

Modern theoretical physics faces a dual crisis. Quantum mechanics splits between deterministic unitary evolution (Schrödinger equation) and stochastic measurement outcomes (Born rule), creating the measurement problem. Cosmology struggles to reconcile geometric expansion (general relativity) with thermodynamic decay (second law). Unistochastic dynamics, introduced by Barandes (1), resolves this by describing systems via stochastic processes in classical-like configuration spaces, governed by transition matrices $\Gamma_{ij}(t) = |U_{ij}(t)|^2$, which replicate quantum phenomena like interference and entanglement without wavefunctions as ontological primitives. The Relativistic Scalar–Vector Plenum (RSVP) extends this to a continuous entropic field, where coarse-graining maps field dynamics to unistochastic matrices:

$$\mathfrak{C}_1 : (\Phi, \mathbf{v}, S) \mapsto \Gamma_{ij}.$$

This monograph argues that ISQM and RSVP are dual scales of a single symplectic substrate, with discrete stochastic transitions coarse-graining into a thermodynamic continuum, providing a geometric foundation for quantum probabilities. Compared to Bohmian mechanics' deterministic non-locality, RSVP's stochastic framework offers a local, thermodynamically consistent ontology.

3 Historical Background: From Unitary Dynamics to Stochastic Ontologies

Quantum mechanics' interpretational challenges have spurred alternatives. Bohmian mechanics (3) posits deterministic trajectories guided by a pilot wave ψ, resolving measurement but introducing non-local influences that violate Bell inequalities non-trivially (2). GRW models (13) add stochastic collapse terms to the Schrödinger equation, breaking unitarity. Everett's many-worlds (14) preserves unitarity via branching realities, at the cost of ontological complexity.

Barandes's ISQM (1) embraces stochasticity, using unistochastic matrices $\Gamma_{ij} = |U_{ij}|^2$ to govern transitions in configuration space. Building on Życzkowski's work on unistochastic matrices (4; 5), ISQM avoids non-locality through non-Markovian history dependence, offering a simpler ontology than Bohmian hidden variables or Everettian branching. The stochastic-quantum correspondence allows Hilbert-space methods as emergent tools, with applications beyond quantum physics, such as turbulence and finance (11).

4 Conceptual Bridge: From Indivisibility to Continuity

ISQM's indivisible transitions are non-factorizable, non-Markovian laws:

$$\Pr(x_{t+\Delta t}|x_t, x_{t-\tau}) \neq \Pr(x_{t+\Delta t}|x_t),$$

breaking the Chapman–Kolmogorov equation, enabling interference and entanglement without amplitudes. A division event, triggered by interactions, resets the history kernel, mimicking decoherence (1). Indivisibility is defined as a non-factorizable transition kernel on a measure space (X, μ), where $\mu(x_i) = \rho_i$, distinguishing it from standard Markov kernels by history dependence. RSVP provides the continuous counterpart, with entropic fields $(\Phi, \mathbf{v}, S)$ whose coarse-graining recovers Γ_{ij}, formalizing a stochastic-quantum correspondence where continuity is the limit of discrete probabilistic acts. Prerequisites include finite-state Markov chains and hyperbolic-parabolic PDE theory.

5 Mathematical Foundations and the Functorial Bridge

5.1 Functional Setting and Existence Theorem

Let Σ be a compact (1+3)-dimensional manifold with boundary $\partial\Sigma$. The RSVP fields:

$$(\Phi, \mathbf{v}, S) : \Sigma \to \mathbb{R} \times T\Sigma \times \mathbb{R}$$

belong to Sobolev spaces $\Phi, S \in H^1(\Sigma)$, $\mathbf{v} \in H^1(\Sigma, \mathbb{R}^3)$. Boundary conditions are:

$$S|_{\partial\Sigma} = 0, \quad (\Phi\mathbf{v} - D\nabla\Phi) \cdot \mathbf{n}|_{\partial\Sigma} = 0.$$

The field equations are:

$$\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = \sigma(S), \tag{1}$$

$$\partial_t S = D\nabla^2 S - \mathbf{v} \cdot \nabla S + f(\Phi), \tag{2}$$

with $\sigma(S) = -\kappa S^2$, $f(\Phi) = \lambda\Phi$, $\kappa, \lambda > 0$, satisfying Lipschitz bounds $|\sigma'(S)| \leq L_\sigma$, $|f'(\Phi)| \leq L_f$.

Theorem 2 (Local Existence and Uniqueness)**.** *For initial data $(\Phi_0, \mathbf{v}_0, S_0) \in H^1(\Sigma) \times H^1(\Sigma, \mathbb{R}^3) \times H^1(\Sigma)$, with $\int_\Sigma \Phi_0 \, d^3x = 1$, there exists $T > 0$ and a unique classical solution $(\Phi, \mathbf{v}, S) \in C([0,T]; H^1(\Sigma))$ to (1)–(2). If $\|\nabla \cdot \mathbf{v}\|_\infty$, $\|\sigma\|_{C^1}$, and $\|f\|_{C^1}$ are bounded uniformly in t, the solution extends globally.*

Proof. The system is a coupled hyperbolic-parabolic PDE. Equation (2) is parabolic in S, with operator $\mathcal{L}_S = \partial_t - D\nabla^2 + \mathbf{v} \cdot \nabla$, generating a C^0 semigroup on $H^1(\Sigma)$ (9). Equation (1) is hyperbolic-conservative in Φ. Rewrite:

$$\Phi(t) = e^{-\int_0^t \nabla \cdot \mathbf{v} \, ds} \Phi_0 + \int_0^t e^{-\int_s^t \nabla \cdot \mathbf{v} \, du} \sigma(S(s)) \, ds.$$

Lipschitz continuity of σ and f yields a contraction mapping in $C([0,T]; H^1(\Sigma))$ for small T. Energy estimates:

$$\frac{d}{dt} \|\Phi\|_{H^1}^2 = 2 \int \Phi \left(\sigma(S) - \nabla \cdot (\Phi \mathbf{v})\right) d^3x \leq C(\|\Phi\|_{H^1}^2 + \|\sigma(S)\|_{H^1}^2),$$

and similarly for S. Grönwall's inequality ensures boundedness. Global extension follows from uniform bounds on $\|\nabla \cdot \mathbf{v}\|_\infty$. Normalization is preserved:

$$\frac{d}{dt} \int \Phi \, d^3x = \int \sigma(S) \, d^3x = -\kappa \int S^2 \, d^3x \leq 0,$$

adjustable via boundary fluxes to maintain $\int \Phi = 1$. □

5.2 Flux Representation and Discrete Cells

Partition Σ into disjoint cells $\{\Omega_i\}_{i=1}^N$. Define the integrated density:

$$\Phi_i = \int_{\Omega_i} \Phi \, d^3x,$$

and flux through shared boundaries:

$$J_{ij} = \int_{\partial\Omega_i \cap \Omega_j} (\Phi \mathbf{v} - D\nabla\Phi) \cdot d\mathbf{A}.$$

By the divergence theorem:

$$\dot{\Phi}_i = \sum_j J_{ij} + \int_{\Omega_i} \sigma(S) \, d^3x.$$

Normalizing by Φ_j:

$$\Gamma_{ij} = \frac{J_{ij}}{\Phi_j}.$$

In the continuum limit $\Delta x, \Delta t \to 0$, Γ_{ij} approximates the operator $\nabla \cdot (D\nabla\Phi - \Phi\mathbf{v})$, yielding the unistochastic kernel.

5.3 Definition of the Coarse-Graining Functor

Let **RSVPField** be the category with objects $(\Phi, \mathbf{v}, S)$ satisfying (1)–(2), and morphisms as smooth entropy-preserving diffeomorphisms $f : \Sigma \to \Sigma$ with $f^*(\Phi, \mathbf{v}, S) = (\Phi', \mathbf{v}', S')$. Let **UniStoch** be the category with objects as doubly stochastic matrices $\Gamma = (\Gamma_{ij})$, and morphisms as stochastic intertwiners Λ satisfying $\Lambda\Gamma = \Gamma'\Lambda$.

[Coarse-Graining Functor] The map:

$$\mathfrak{C}_1 : \mathbf{RSVPField} \to \mathbf{UniStoch}, \quad \mathfrak{C}_1(\Phi, \mathbf{v}, S) = (\Gamma_{ij})_{i,j} = \left(\frac{1}{\Phi_j} \int_{\partial\Omega_i \cap \Omega_j} (\Phi\mathbf{v} - D\nabla\Phi) \cdot d\mathbf{A} \right)_{i,j}$$

is the coarse-graining functor.

Proposition 1 (Functorial Properties). $\mathfrak{C}_1$ *preserves identities and composition:*

$$\mathfrak{C}_1(\mathrm{id}) = I, \quad \mathfrak{C}_1(F_{t_2} \circ F_{t_1}) = \mathfrak{C}_1(F_{t_2})\mathfrak{C}_1(F_{t_1}).$$

Proof. For the identity flow $F_t = \mathrm{id}$, $\mathbf{v} = 0$, so $J_{ij} = 0$ for $i \neq j$, and $\Gamma_{ij} = \delta_{ij}$. For sequential flows $F_{t_2} \circ F_{t_1}$, consider the flux over intermediate cells:

$$J_{ij}(t_1 + t_2) = \int_0^{t_1+t_2} \int_{\partial\Omega_i \cap \Omega_j} (\Phi\mathbf{v} - D\nabla\Phi) \cdot d\mathbf{A}\, dt.$$

Splitting at t_1, the composition of flows corresponds to matrix multiplication $\Gamma(t_2)\Gamma(t_1)$, as intermediate boundary integrals cancel by conservation. The adjoint $\mathfrak{R}_1 : \mathbf{UniStoch} \to \mathbf{RSVPField}$ reconstructs fields, with unit $\eta : \mathrm{Id} \to \mathfrak{R}_1 \circ \mathfrak{C}_1$ as the discretization map and counit $\epsilon : \mathfrak{C}_1 \circ \mathfrak{R}_1 \to \mathrm{Id}$ as the smoothing limit, satisfying $d\eta = d\epsilon = dS$. □

5.4 Entropy-Preserving Refinement

The adjoint functor $\mathfrak{R}_1$ assigns to each Γ a smooth field configuration reproducing the same transition probabilities. The homotopy-adjunction $\mathfrak{C}_1 \dashv \mathfrak{R}_1$ ensures:

$$\mathrm{Hom}_{\mathbf{RSVPField}}(\mathfrak{R}_1\Gamma, \Phi) \simeq \mathrm{Hom}_{\mathbf{UniStoch}}(\Gamma, \mathfrak{C}_1\Phi).$$

Non-uniqueness in $\mathfrak{R}_1$ reflects gauge freedom in $\mathbf{v}$, addressed by fixing $\nabla \cdot \mathbf{v} = 0$.

6 Explicit Unitary Reconstruction and Numerical Example

6.1 From Entropic Flux to Unistochastic Matrices

Given a discretized RSVP field on $\{\Omega_i\}$, the transition coefficients are:

$$\Gamma_{ij} = \frac{1}{\Phi_j} \int_{\partial\Omega_i \cap \Omega_j} (\Phi \mathbf{v} - D\nabla\Phi) \cdot d\mathbf{A}.$$

For fine discretizations, Γ is doubly stochastic:

$$\sum_i \Gamma_{ij} = 1, \quad \sum_j \Gamma_{ij} = 1, \quad \Gamma_{ij} \geq 0.$$

6.2 Local Phase Potential

Let the lamphrodic potential $\theta(x)$ satisfy $\mathbf{v} = \nabla\theta$. Define cell-averaged phases:

$$\theta_i = \frac{1}{\mathrm{Vol}(\Omega_i)} \int_{\Omega_i} \theta(x)\, d^3x.$$

The phase difference $\theta_{ij} = \theta_i - \theta_j$ generates:

$$U_{ij} = \sqrt{\Gamma_{ij}}\, e^{i\theta_{ij}/\hbar}.$$

Unitarity requires:

$$\sum_j U_{ij} U_{kj}^* = \delta_{ik}, \quad \sum_j \sqrt{\Gamma_{ij}\Gamma_{kj}}\, e^{i(\theta_{ij}-\theta_{kj})/\hbar} = 0 \text{ for } i \neq k.$$

Theorem 3 (Unitary Embedding via Lamphrodic Potential)**.** *If Γ is doubly stochastic and θ satisfies the orthogonality condition, then $U_{ij} = \sqrt{\Gamma_{ij}} e^{i\theta_{ij}/\hbar}$ is unitary, and $\Gamma_{ij} = |U_{ij}|^2$ is unistochastic. Any unistochastic Γ admits such a potential up to gauge transformations $\theta_i \mapsto \theta_i + 2\pi k_i \hbar$.*

Proof. Given $\Gamma_{ij} = |U_{ij}|^2$, unitarity of U ensures $\sum_i |U_{ij}|^2 = 1$, matching double-stochasticity. Construct $U_{ij} = \sqrt{\Gamma_{ij}} e^{i\theta_{ij}/\hbar}$, where $\theta_{ij} = \theta_i - \theta_j$. The orthogonality condition:

$$\sum_j \sqrt{\Gamma_{ij}\Gamma_{kj}} e^{i(\theta_{ij}-\theta_{kj})/\hbar} = 0,$$

ensures $U^\dagger U = I$. Conversely, for any unistochastic Γ, set $\theta_{ij} = \hbar \arg(U_{ij})$, with $\mathbf{v} = \nabla\theta$. Gauge transformations $\theta_i \mapsto \theta_i + 2\pi k_i \hbar$ leave U invariant up to a diagonal phase, preserving Γ. Existence follows from Życzkowski's results (4), as doubly stochastic matrices in the unitary convex hull admit such factorization. □

6.3 Example: Two-Cell Entropic Exchange

For a 1D plenum with two cells Ω_1, Ω_2, width $\Delta x = 1$, set $\Phi_1 = \Phi_2 = 1$, $\mathbf{v} = (v, 0, 0)$, $v = 2$, $D = 0.1$, $\Delta t = 0.1$. The flux is:

$$J_{12} = J_{21} = v\Delta t = 0.2, \quad \Gamma = \begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{pmatrix}.$$

Verify unitarity with:

$$U = \begin{pmatrix} \sqrt{0.8} & \sqrt{0.2} \\ \sqrt{0.2} & -\sqrt{0.8} \end{pmatrix} \approx \begin{pmatrix} 0.894 & 0.447 \\ 0.447 & -0.894 \end{pmatrix}.$$

Check: $U^\dagger U = I$. Eigenvalues are ± 1, confirming unitarity.

6.4 Example: Three-Cell Circular Flow

For three cells connected cyclically, flux $J = 0.1$:

$$\Gamma = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.8 & 0.1 \\ 0.1 & 0.1 & 0.8 \end{pmatrix}.$$

Set $\theta_{ij} = 2\pi(i - j)/3$, so:

$$U_{ij} = \sqrt{\Gamma_{ij}} e^{2\pi i(i-j)/3}.$$

Eigenvalues are $1, e^{2\pi i/3}, e^{-2\pi i/3}$, confirming unistochasticity.

6.5 Numerical Verification Procedure

1. Compute Γ from fluxes. 2. Verify $\sum_i \Gamma_{ij} = \sum_j \Gamma_{ij} = 1$. 3. Solve for θ_{ij} satisfying orthogonality. 4. Construct $U_{ij} = \sqrt{\Gamma_{ij}} e^{i\theta_{ij}/\hbar}$. 5. Check $\|U^\dagger U - I\| < \epsilon$.

6.6 Interpretation

Γ quantifies entropic exchange, with θ encoding coherence. Microscopically, these are ISQM's indivisible events; macroscopically, they form RSVP's continuum.

7 Derived Quantization Example: Boundary Emergence of the Unistochastic Kernel

7.1 AKSZ Background

The AKSZ construction maps a d-dimensional source supermanifold $T[1]\Sigma$ and a target n-shifted symplectic manifold $(\mathcal{X}, \omega_n, \Theta)$ to a BV action:

$$S_{\text{AKSZ}} = \int_{T[1]\Sigma} \left(\langle \Phi^*(\alpha), d\Phi \rangle + \Phi^*(\Theta) \right),$$

where α is the Liouville 1-form, Θ satisfies $\{\Theta, \Theta\} = 0$. For RSVP, $\mathcal{X} = \mathbf{T}^*[0]X$, with $\omega_0 = \delta\Phi \wedge \delta^*(\mathbf{v})$, and:

$$\Theta = \iota_{\mathbf{v}}(\omega_0) + Dg^{ab}\partial_a S \partial_b S.$$

7.2 RSVP AKSZ Action

The action is:

$$S_{\mathrm{AKSZ}} = \int_{T[1]\Sigma} \left(\Phi dS + \mathbf{v} \cdot \nabla S - Dg^{ab}\partial_a S \partial_b S\right).$$

Integrating over odd coordinates yields:

$$\mathcal{L}_{\mathrm{RSVP}} = \Phi(\partial_t S + \mathbf{v} \cdot \nabla S) - D|\nabla S|^2.$$

7.3 Path Integral and Boundary Reduction

Quantization uses:

$$Z = \int [\mathcal{D}\Phi][\mathcal{D}\mathbf{v}][\mathcal{D}S] \exp\left(\frac{i}{\hbar}\int_\Sigma \mathcal{L}_{\mathrm{RSVP}}\right).$$

Fix boundary data $(\Phi, S)|_{\partial\Sigma}$. The boundary current is:

$$J^\mu = \Phi\partial^\mu S - D\partial^\mu(\Phi S), \quad \delta S_{\mathrm{AKSZ}}|_{\partial\Sigma} = \int_{\partial\Sigma} J^\mu \delta S \, dA_\mu.$$

The boundary path integral yields:

$$Z_{\partial\Sigma}[\Phi_\partial, S_\partial] = \exp\left(\frac{i}{\hbar}\int_{\partial\Sigma} \Phi_\partial \Delta S_\partial\right).$$

7.4 Discretization and Emergent Transition Kernel

Partition $\partial\Sigma$ into cells $\{\Omega_i\}$. The boundary action is:

$$S_\partial = \sum_{i,j} \Phi_i \left(\Delta t \mathbf{v}_{ij} \cdot \nabla S_j - D\Delta t |\nabla S_j|^2\right).$$

The transition amplitude is:

$$A_{ij} = \exp\left[\frac{i}{\hbar}\left(\Delta t \mathbf{v}_{ij} \cdot \nabla S_j - D\Delta t|\nabla S_j|^2\right)\right], \quad U_{ij} = \frac{1}{\sqrt{\mathcal{N}}} A_{ij},$$

with $\mathcal{N} = \sum_{ij} |A_{ij}|^2$. The kernel is:

$$\Gamma_{ij} = |U_{ij}|^2 = \frac{1}{\mathcal{N}} \exp\left[-\frac{2D\Delta t}{\hbar}|\nabla S_j|^2\right].$$

7.5 Semiclassical Expansion

Expanding around S_0:

$$S = S_0 + \hbar^{1/2}\delta S + O(\hbar),$$

the one-loop correction is:

$$\Gamma_{ij}^{(1)} = \Gamma_{ij}^{(0)} \left[1 - \frac{\hbar D\Delta t}{2}(\nabla^2 S_0)^2 + O(\hbar^2)\right].$$

7.6 Interpretation

The unistochastic kernel emerges as the boundary manifestation of RSVP's quantization, with quantum stochasticity as discrete sampling of entropic geometry.

8 Entropy–Curvature Coupling and the Born Rule as a Limit Theorem

8.1 Entropy–Curvature Interaction

On a curved Σ with scalar curvature R, the Lagrangian includes:

$$\mathcal{L}_{\text{curv}} = \alpha_1 R S^2 + \alpha_2 R_{\mu\nu} \nabla^\mu S \nabla^\nu S + \alpha_3 R |\nabla S|^2.$$

The total Lagrangian is:

$$\mathcal{L}_{\text{tot}} = \Phi(\partial_t S + \mathbf{v} \cdot \nabla S) - D|\nabla S|^2 + \mathcal{L}_{\text{curv}}.$$

8.2 Curvature Correction to the Transition Kernel

The curvature-corrected kernel is:

$$\Gamma_{ij}^{(\text{curv})} = |U_{ij}|^2 \exp\left[-\frac{2D\Delta t}{\hbar}|\nabla S_j|^2 - \hbar\left(\alpha_1 R S_j^2 + \alpha_3 R|\nabla S_j|^2\right)\right].$$

8.3 Long-Time Limit and Ergodic Averaging

The probability evolves:

$$\rho_i(t + \Delta t) = \sum_j \Gamma_{ij}^{(\text{curv})} \rho_j(t).$$

For large N:

$$\rho_i(t + N\Delta t) = (\Gamma^{(\text{curv})})_{ij}^N \rho_j(t_0).$$

By the Perron–Frobenius theorem, $\rho_i \to \rho_i^{(\text{eq})}$, satisfying:

$$\rho_i^{(\text{eq})} = \sum_j |U_{ij}|^2 \rho_j^{(\text{eq})}.$$

Theorem 4 (Born Rule as Entropic Limit)**.** *For an ergodic* $\Gamma^{(\mathrm{curv})}$*, empirical frequencies* $f_i^{(N)} = \frac{1}{N}\sum_{k=1}^{N}\chi_i(t_k)$ *converge almost surely to* $\rho_i^{(\mathrm{eq})} = \sum_j |U_{ij}|^2\rho_j^{(\mathrm{eq})}$*.*

Proof. Since $\Gamma^{(\mathrm{curv})}$ is stochastic and ergodic, the Markov chain satisfies Birkhoff's ergodic theorem:

$$\lim_{N\to\infty}\frac{1}{N}\sum_{k=1}^{N}\Gamma^k = \Pi,$$

where Π projects onto the stationary subspace. As $\Gamma_{ij} = |U_{ij}|^2$, the stationary distribution obeys the Born rule. Convergence follows from the strong law of large numbers for Markov chains (9). □

8.4 Interpretation

The Born rule is a statistical attractor of RSVP dynamics, with curvature accelerating equilibration.

9 Cosmological Extension: The Unistochastic Horizon and Expyrotic Reset

9.1 From Local Indivisibility to Cosmic Smoothing

Microscopic indivisible transitions (Γ_{ij}) aggregate into RSVP's entropic flow. The cosmic horizon is $\partial\Sigma$, where local transitions become global thermodynamic gradients.

9.2 Entropy Production and Horizon Geometry

Total entropy is:

$$S_{\mathrm{tot}}(t) = \int_\Sigma \Phi S\, d^3x, \quad \frac{dS_{\mathrm{tot}}}{dt} = \int_\Sigma \sigma(S)\, d^3x = -\kappa\int S^2\, d^3x \leq 0.$$

The entropic Hubble parameter is:

$$H_{\mathrm{RSVP}} = \frac{1}{3}\nabla\cdot\mathbf{v}.$$

9.3 The Unistochastic Horizon

The horizon kernel is:

$$\Gamma_{ij}^{(\cos)} = |U_{ij}|^2\exp\left[-\frac{\Lambda_{\mathrm{ent}}d_{ij}^2}{\hbar c}\right].$$

9.4 Entropy–Curvature Feedback and the Expyrotic Limit

Curvature reduces as:

$$\nabla S \to 0, \quad R \to 0, \quad H_{\text{RSVP}} \to 0,$$

approaching Expyrosis.

9.5 Quantum Seeds and the Reset Mechanism

At Expyrosis, $\Gamma_{ij} \to \frac{1}{N}$. Fluctuations ϵ_{ij} grow as:

$$\epsilon_{ij}(t) \sim e^{\lambda t}, \quad \lambda = 2\kappa S.$$

9.6 Cosmological Born Rule

Macrostate probabilities follow:

$$P(\mathcal{O}_i) = \lim_{T\to\infty} \frac{1}{T} \int_0^T |U_{ij}(t)|^2 \, dt.$$

9.7 Interpretation

The universe equilibrates through entropic cycles, with stochastic resets seeding new structure.

9.8 Summary

- The horizon is the boundary of coherence. - Curvature drives Expyrosis. - Fluctuations trigger new cycles. - The Born rule extends to cosmological scales.

10 Unified View: Indivisibility Across Scales

10.1 Indivisibility as a Universal Principle

Indivisibility resists arbitrary partition. Microscopically, it is the unistochastic kernel; mesoscopically, RSVP's smooth fields; cosmologically, the coherent plenum.

10.2 Scale Hierarchy of the Plenum

Scale	**Dominant Structure**	**Conservation Law**
Quantum	Unistochastic kernel $\lvert U_{ij}\rvert^2$	Unitarity (phase coherence)
Mesoscopic	RSVP field $(\Phi, \mathbf{v}, S)$	Entropy continuity
Macroscopic	Entropy curvature R, S	Thermodynamic balance
Cosmic	Horizon kernel $\Gamma_{ij}^{(\cos)}$	Global entropy conservation

10.3 The RSVP–UQM Correspondence Principle

The functor $\mathfrak{C}_1 : \textbf{RSVPField} \to \textbf{UniStoch}$ and adjoint $\mathfrak{R}_1$ define:

$$\mathfrak{C}_1(\Phi, \mathbf{v}, S) = |U|^2, \quad \mathfrak{R}_1(|U|^2) = (\Phi, \mathbf{v}, S).$$

10.4 From Quantization to Smoothing

The deformation parameters are:

$$\hbar \text{ (quantum)}, \quad D \text{ (smoothing)}, \quad \Lambda_{\text{RSVP}} = \frac{D}{\hbar}.$$

10.5 Expyrotic Cycles as Indivisible Renewal

Expyrotic resets re-quantize the universe via stochastic fluctuations.

10.6 Philosophical Implications

1. Reality as recursive coherence. 2. Measurement as entropic coupling. 3. Time as smoothing gradient. 4. Cosmos as indivisible field.

10.7 Outlook

- Derived Markov stacks. - Numerical RSVP simulations. - Cosmological data fitting.

10.8 Closing Reflection

The universe equilibrates, not expands. The wave smooths, not collapses. Reality reconstitutes, not divides.

The universe does not expand; it equilibrates. The wave does not collapse; it smooths. Reality does not divide; it reconstitutes.

A Derived–Geometric Form of the RSVP Field

A.1 The Graded Manifold

Let Σ be the (1+3)-dimensional worldvolume, and $T[1]\Sigma$ its parity-shifted tangent bundle with coordinates (x^μ, θ^μ), degrees $(0, 1)$. The RSVP fields define $\Phi : T[1]\Sigma \to \mathcal{X}$, where $\mathcal{X} = \mathbf{T}^*[0]X \simeq \operatorname{Spec}\operatorname{Sym}(\mathbb{T}_X[0])$, with:

$$\omega_0 = \int_\Sigma \delta\Phi \wedge \delta^*(\mathbf{v}).$$

A.2 The AKSZ Action

$$S_{\text{AKSZ}} = \int_{T[1]\Sigma} \left(\Phi dS + \mathbf{v} \cdot \nabla S - D g^{ab} \partial_a S \partial_b S \right).$$

The Hamiltonian $\Theta = \iota_{\mathbf{v}}(\omega_0) + D g^{ab} \partial_a S \partial_b S$ satisfies $\{\Theta, \Theta\} = 0$.

Proof. The Poisson bracket is:

$$\{\Theta, \Theta\} = 2\iota_{\mathbf{v}} d\Theta = 2\iota_{\mathbf{v}} d(\iota_{\mathbf{v}}\omega_0 + D g^{ab} \partial_a S \partial_b S).$$

Since $\mathcal{L}_{\mathbf{v}}\omega_0 = dS$, and $d(g^{ab}\partial_a S \partial_b S) = 0$ for a closed form, the bracket vanishes (7). □

A.3 BV Extension and Master Equation

With antifields $\Phi^*, \mathbf{v}^*, S^*$ (degree -1) and ghosts c^μ, η (degree $+1$):

$$S_{\text{BV}} = S_{\text{AKSZ}} + \int_\Sigma \left[\Phi^*(c^\mu \partial_\mu \Phi) + S^*(c^\mu \partial_\mu S + \eta) + v_i^*(c^\mu \partial_\mu v^i - v^j \partial_j c^i) + \frac{1}{2} c_\mu^* c^\nu \partial_\nu c^\mu + \eta^*(c^\mu \partial_\mu \eta) \right].$$

The master equation $(S_{\text{BV}}, S_{\text{BV}}) = 0$ holds.

Proof. The antibracket is:

$$(F, G) = \int \frac{\delta F}{\delta \Psi^A} \frac{\delta G}{\delta \Psi_A^*} - \frac{\delta F}{\delta \Psi_A^*} \frac{\delta G}{\delta \Psi^A} \, d^4x.$$

Compute $(S_{\text{BV}}, S_{\text{BV}})$: - AKSZ terms: $\{\Theta, \Theta\} = 0$. - Ghost terms: Diffeomorphism and gauge algebras close. - Antifield terms: Cancel via integration by parts, as $\partial_\mu(c^\mu \partial_\nu c^\nu) = 0$. Ghost number conservation ensures all terms balance (7). □

A.4 Derived Quantization and Boundary Data

Quantization maps to $\text{Crit}(S_{\text{BV}})$. Boundary restriction yields:

$$Z_{\partial\Sigma}[\Gamma] = \int_{\Phi|_{\partial\Sigma} = \Gamma} e^{\frac{i}{\hbar} S_{\text{AKSZ}}}.$$

A.5 Interpretation

The AKSZ–BV framework unifies unitarity and entropy production.

B BV–BRST Structure of the RSVP–AKSZ Sigma Model

B.1 Field Content and Degrees

$$\begin{array}{rcccl} \Phi & : & \Sigma[0] & \to & \text{Scalar entropy density} \\ \mathbf{v}_i & : & \Sigma[0] & \to & \text{Vector flow field} \\ S & : & \Sigma[0] & \to & \text{Entropy potential} \\ c^\mu & : & \Sigma[1] & \to & \text{Diffeomorphism ghost} \\ \eta & : & \Sigma[1] & \to & \text{Entropy–gauge ghost} \end{array}$$

Ghost numbers: $\mathrm{gh}(c^\mu) = 1$, $\mathrm{gh}(\eta) = 1$, $\Phi, \mathbf{v}, S$ at 0. Antifields Ψ^*_A have degree $-1 - \mathrm{gh}(\Psi^A)$.

B.2 Classical Action in BV Form

$$S_{\mathrm{BV}} = \int_\Sigma \Big[\Phi(\partial_t S + \mathbf{v} \cdot \nabla S) - D|\nabla S|^2 + \Phi^*(c^\mu \partial_\mu \Phi) + S^*(c^\mu \partial_\mu S + \eta) + v_i^*(c^\mu \partial_\mu v^i - v^j \partial_j c^i) + \frac{1}{2} c^*_\mu c^\nu \partial_\nu c^\mu +$$

B.3 BRST Transformations

$$s\Phi = c^\mu \partial_\mu \Phi, \quad sS = c^\mu \partial_\mu S + \eta, \quad sv^i = c^\mu \partial_\mu v^i - v^j \partial_j c^i, \quad sc^\mu = c^\nu \partial_\nu c^\mu, \quad s\eta = c^\mu \partial_\mu \eta.$$

These satisfy $s^2 = 0$ up to equations of motion.

B.4 Classical Master Equation

$$(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0.$$

B.5 Observables and Cohomology

Observables like $\mathcal{O}_f = \int_\Sigma f(\Phi, S) d^3x$ lie in BV cohomology.

B.6 Interpretation

Ghosts encode symmetries, antifields enforce conservation.

C Derived Quantization and Emergent Probability Law

C.1 BV Path Integral and Gauge Fixing

$$Z = \int [\mathcal{D}\Psi][\mathcal{D}\Psi^*] \exp\left(\frac{i}{\hbar} S_{\mathrm{BV}}\right).$$

Gauge fixing imposes $G^\mu = \partial_\nu v^\nu$, $G_\eta = \partial_t S$.

C.2 Integration over Ghosts and Antifields

$$Z = \int [\mathcal{D}\Phi][\mathcal{D}\mathbf{v}][\mathcal{D}S] \exp\left(\frac{i}{\hbar}\int_\Sigma \mathcal{L}_{\mathrm{RSVP}}\right) \times \det(\Delta_{\mathrm{FP}}).$$

C.3 Emergent Unistochastic Kernel

$$Z \simeq \sum_{\{\Phi_i(t)\}} \exp\left[\frac{i}{\hbar}\sum_{t,i,j} \Phi_i(t)\Gamma_{ij}S_j(t+\Delta t)\right],$$

with $\Gamma_{ij} = \frac{1}{N}\left|\exp\left(i\Delta t\mathbf{v}\cdot\nabla S - D\Delta t|\nabla S|^2\right)\right|^2$.

C.4 Born Rule as a Limit Theorem

$$\lim_{N\to\infty}\frac{N_i(t)}{N} = \sum_j \Gamma_{ij}\rho_j(t) = |U_{ij}|^2\rho_j(t).$$

C.5 Derived Stack Quantization

The critical locus $\mathrm{Crit}(S_{\mathrm{BV}})$ defines a (-1)-shifted symplectic stack.

C.6 Interpretation

The BV quantization bridges micro-unitarity and macro-entropy.

D Semiclassical Expansion and Entropy–Curvature Anomalies

D.1 Semiclassical Expansion

Expanding around Ψ_0:

$$\Gamma^{(1)} = S_{\mathrm{RSVP}}[\Psi_0] + \frac{i\hbar}{2}\ln\det\mathcal{K}_{AB} + O(\hbar^2).$$

The fluctuation operator is:

$$\mathcal{K}_{AB} = \frac{\delta^2 S_{\mathrm{RSVP}}}{\delta\Psi^A\delta\Psi^B}.$$

Proof. Compute $\mathcal{K}$:

$$\mathcal{K}_{\Phi S} = \partial_t S + \mathbf{v}\cdot\nabla S, \quad \mathcal{K}_{SS} = -D\nabla^2\Phi.$$

The determinant is regularized via ζ-function, yielding curvature terms (6). □

D.2 Curvature Coupling and Entropic Torsion

$$\Gamma_{\text{anom}} = \frac{\hbar}{24\pi^2} \int_\Sigma \left(\alpha_1 R \Phi_0 + \alpha_2 T^{\rho}_{\mu\nu} \mathbf{v}_0^{\mu} \nabla_\rho S_0 + \alpha_3 |\nabla S_0|^2 R \right) d^4x.$$

D.3 Effective Unistochastic Kernel

$$\Gamma^{(\text{eff})}_{ij} = |U_{ij}|^2 \exp\left[-\hbar \mathcal{A}_{ij}(\Phi_0, S_0, R, T) \right].$$

D.4 Shifted Symplectic Anomaly Form

$$d\omega_0 = \hbar \Omega^{(1)}, \quad \Omega^{(1)} = \mathrm{Tr}\left(\mathcal{R} \wedge \mathcal{R} + \mathcal{T} \wedge \mathcal{T} \right).$$

D.5 Physical Interpretation

Anomalies introduce entropy leakage.

D.6 Summary

The expansion unifies classical entropy with quantum corrections.

E Renormalization–Group Flow of Entropy and Curvature

E.1 Scale Dependence

$$\mu \frac{d\Gamma}{d\mu} = \left(\beta_D \frac{\partial}{\partial D} + \beta_{\alpha_i} \frac{\partial}{\partial \alpha_i} - \gamma_\Phi \int \Phi \frac{\delta}{\delta \Phi} - \gamma_S \int S \frac{\delta}{\delta S} \right) \Gamma.$$

E.2 One–Loop Beta Functions

$$\beta_D = \frac{\hbar}{6\pi^2} D^2 \left(1 - \frac{5}{12}\epsilon \right) - \alpha_3 R D, \quad \beta_{\alpha_1} = \frac{\hbar}{24\pi^2}(1-6D), \quad \beta_{\alpha_2} = \frac{\hbar}{8\pi^2} D, \quad \beta_{\alpha_3} = \frac{\hbar}{12\pi^2}(1-3D).$$

Proof. Using dimensional regularization ($d = 4 - \epsilon$), compute one-loop diagrams for Φ, S propagators. The β_D term arises from the vertex $\Phi \mathbf{v} \cdot \nabla S$, with logarithmic divergences proportional to D^2. Curvature terms couple via $\alpha_3 R$, stabilizing the flow (12). □

E.3 Fixed Points

- **Microscopic**: $D \to 0$, unitary. - **Mesoscopic**: $D_* = 6\pi^2 \hbar^{-1} \alpha_3 R$, RSVP. - **Macroscopic**: D grows, Expyrotic.

Proof. Solve $\beta_D = 0$:

$$\frac{\hbar}{6\pi^2} D^2 \left(1 - \frac{5}{12}\epsilon\right) - \alpha_3 R D = 0.$$

Stable fixed point at $D_* = 6\pi^2\hbar^{-1}\alpha_3 R$. Jacobian $\partial\beta_D/\partial D$ has negative eigenvalues at D_* (12). □

E.4 Flow of Entropy–Curvature Coupling

$$\mu \frac{d\Lambda_{\text{ent}}}{d\mu} = (\beta_{\alpha_1} R + \beta_{\alpha_3} |\nabla S|^2) + (\alpha_1 + \alpha_3)\beta_R.$$

E.5 Geometric Interpretation

The RG flow traces a homotopy on ω_0.

E.6 Physical Picture

D and α_i drive the plenum from unitarity to smoothness.

E.7 Summary

The RG flow unifies scales, with fixed points marking universality classes.

F Phase–Space Portrait and Entropic Gradient Flow

F.1 Entropic Potential

$$\mathcal{S}[\Phi, S] = \int_\Sigma \left(\Phi S - \frac{1}{2} D |\nabla S|^2 + \frac{1}{2}\alpha_1 R S^2 + \frac{1}{2}\alpha_3 R |\nabla S|^2\right) d^4x.$$

F.2 Phase–Space Variables

Coordinates $(q^1, q^2, q^3) = (\Phi, S, D)$, momenta $(p_1, p_2, p_3) = (\nabla \cdot \mathbf{v}, \partial_t S, \partial_t D)$, with $\Omega = \sum_i dq^i \wedge dp_i$.

F.3 Entropic Phase Portrait

$$\frac{d}{d\tau}\begin{pmatrix}\delta D \\ \delta\alpha_i\end{pmatrix} = J \begin{pmatrix}\delta D \\ \delta\alpha_i\end{pmatrix}.$$

Fixed points: UV-attractive (unistochastic), marginal (RSVP), IR-attractive (Expyrotic).

F.4 Derived Gradient Flow

$$\iota_{Q_{\text{RG}}} \omega_{\text{RG}} = d\mathcal{S}.$$

F.5 Cosmological and Informational Outlook

The flow defines a thermodynamic arrow, with D as the learning rate.

F.6 Summary

The portrait unifies regimes via entropic gradient descent.

G Categorical Synthesis of the RSVP–UQM Hierarchy

G.1 Overview

$\mathbf{UniStoch}[r, shiftleft = .7ex, "\mathfrak{C}_1"]\mathbf{RSVPField}[r, shiftleft = .7ex, "\mathfrak{C}_2"][l, shiftleft = .7ex, "\mathfrak{R}_1"]\mathbf{BVQ}$

G.2 Layered Structure

Level	Mathematical Object	Structure Preserved	Physical Interpretation
I. Unistochastic	$\Gamma_{ij} = \lvert U_{ij}\rvert^2$	Doubly-stochastic	Stochastic microdynamics
II. RSVP Field	$(\Phi, \mathbf{v}, S)$	$\nabla \cdot \mathbf{v} = 0$	Entropic continuum
III. BV–AKSZ	$T^*[-1]\mathcal{F}_{\mathrm{RSVP}}$	0-shifted symplectic	Quantized entropy
IV. RG Flow	$(\mathcal{S}, Q_{\mathrm{RG}})$	Gradient descent	Scale-dependent smoothing
V. Global Stack	$\mathcal{M}_{\mathrm{RSVP}}$	Symplectic homotopy	Unified entropy–geometry

G.3 Derived–Stack Composition

$$\mathfrak{Q}_{\mathrm{RSVP}} = \mathfrak{C}_3 \circ \mathfrak{C}_2 \circ \mathfrak{C}_1.$$

G.4 Homotopy–Commutative Cube

$[-2em]\mathcal{F}_{\mathrm{RSVP}}[dr, "\text{Quantize}"][dl, "\text{Coarse-grain}"]\mathcal{U}_{\mathrm{unistoch}}[rr, "\text{Derived BV}"]\mathcal{Q}_{\mathrm{BV/AKSZ}}[dl, "\text{RG flow}"]\mathcal{R}_{\mathrm{RG}}[u$

G.5 Monoidal and Probabilistic Structure

Monoidal products compose subsystems; $\mathbb{P} : \mathbf{UniStoch} \to \mathbf{Set}$ is lax monoidal.

G.6 Summary

The hierarchy forms a derived adjoint tower.

H Categorical Formulation of the RSVP–UQM Tower

H.1 Categories and Objects

- **UniStoch**: Kernels Γ_{ij}, stochastic intertwiners. - **RSVPField**: $(\Phi, \mathbf{v}, S)$, entropy-preserving diffeomorphisms. - **BVQuant**: $(\mathcal{F}, \omega_0, \Theta)$, symplectic quasi-isomorphisms. - **RGFlow**: $(\mathcal{S}, Q_{\mathrm{RG}})$, scale-preserving homotopies.

H.2 Functorial Relationships

$\mathfrak{C}_i$ coarse-grain, $\mathfrak{R}_i$ refine, forming homotopy-adjunctions.

H.3 Natural Transformations

η_i, ϵ_i satisfy $d\eta_i = d\epsilon_i = dS$.

H.4 Global Derived Stack

$$\mathcal{M}_{\mathrm{RSVP}} = \operatorname{hocolim} \mathcal{D}.$$

H.5 Monoidal and Probabilistic Structure

Monoidal products ensure consistency.

H.6 Final Summary

The RSVP–UQM theory unifies stochasticity and geometry.

I Entropy as the Geometry of Indivisibility

I.1 Thermodynamic Completion of Quantum Mechanics

RSVP embeds quantum amplitudes in a geometric medium, resolving measurement issues locally, unlike Bohmian non-locality.

I.2 Relation to Statistical and Information Theories

Links to Jaynes's maximum entropy (17) and information geometry.

I.3 Comparison with Other Interpretations

Feature	ISQM	RSVP	Bohm	GRW
Ontology	Stochastic	Entropic field	Deterministic	Stochastic collapse
Non-locality	No	No	Yes	No
Measurement	Intrinsic jumps	Entropic smoothing	Pilot wave	Collapse terms
Born Rule	Emergent	Ergodic limit	Derived	Postulated

I.4 Numerical Methods for RSVP Simulation

Discretize Σ, solve (1)–(2) using finite difference methods, compute Γ_{ij}.

I.5 Observable Predictions

RSVP predicts curvature-dependent decoherence rates, testable in quantum optics.

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